EXERCISE-I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION

1. $a \neq b \neq c$, $\alpha = \gamma = 90° \beta \neq 90°$ represents :
(1) tetragonal system
(2) orthorhombic system
(3) monoclinic system
(4) triclinic system

SS0001

2. The most unsymmetrical crystal system is :
(1) Cubic (2) Hexagonal
(3) Triclinic (4) Orthorhombic

SS0002

3. Bravais lattices are of :
(1) 10 types (2) 8 types
(3) 7 types (4) 14 types

SS0003

4. The crystal system of a compound with unit cell dimensions a = 0.387, b = 0.387 and c=0.504 nm and $\alpha=\beta$ =90° and γ=120° is :
(1) Cubic (2) Hexagonal
(3) Orthorhombic (4) Rhombohedral

SS0004

ANALYSIS OF CUBIC CRYSTAL

5. In a simple cubic cell, each point on a corner is shared by :
(1) 2 unit cells (2) 1 unit cell
(3) 8 unit cells (4) 4 unit cells

SS0005

6. In a face centred cubic cell, an atom at the face contributes to the unit cell :
(1) 1 part (2) 1/2 part
(3) 1/4 part (4) 1/8 part

SS0006

7. In a body centred cubic cell, an atom at the body centre is shared by :
(1) 1 unit cell (2) 2 unit cell
(3) 3 unit cells (4) 4 unit cells

SS0007

8. Which of the following type of cubic lattice has maximum number of atoms per unit cell ?
(1) Simple cubic
(2) Body centred cubic
(3) Face centred cubic
(4) All have same

SS0008

9. The number of atoms present in a unit cell of a monoatomic element of a simple cubic lattice, body-centred cubic and face centred cubic respectively are
(1) 8, 9 and 14 (2) 1, 2 and 4
(3) 4, 5 and 6 (4) 2, 3 and 5

SS0009

10. Which one of the following is a primitive unit cell ?
(1) Simple cubic (2) BCC
(3) FCC (4) BCC and FCC both

SS0010

11. In a body centred cubic unit cell, a metal atom at the centre of the cell is surrounded by how many other metal atoms :
(1) 8 (2) 6 (3) 12 (4) 4

SS0011

12. A compound is formed by elements A and B. This crystallises in the cubic structure when atoms A are at the corners of the cube and atoms B are at the centre of the body. The simplest formula of the compound is :
(1) AB (2) AB_2 (3) A_2B (4) AB_4

SS0012

13. The fraction of total volume occupied by the atoms present in a simple cube is –

(1) $\frac{\pi}{6}$ (2) $\frac{\pi}{3\sqrt{2}}$ (3) $\frac{\pi}{4\sqrt{2}}$ (4) $\frac{\pi}{4}$

SS0083

14. Sodium metal crystallises in BCC lattice with the cell edge length (a) = 42.29 Å. What is the radius of sodium atom ?
(1) 1.86 Å (2) 1.90 Å (3) 18.3 Å (4) 1.12 Å

SS0014

15. An element has BCC structure having unit cells 12.08×10^{23}. The number of atoms in these cells is :
(1) 12.08×10^{23} (2) 24.16×10^{23}
(3) 48.38×10^{23} (4) 12.08×10^{22}

SS0015

16. A metal has BCC structure and the edge length of its unit cell is 3.04 Å. The volume of the unit cell in cm^3 will be :
(1) 1.6×10^{-21} cm^3 (2) 2.81×10^{-23} cm^3
(3) 6.02×10^{-23} cm^3 (4) 6.6×10^{-24} cm^3

SS0016

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CRYSTAL DENSITY

17. If 'a' stands for the edge length of the cubic systems: simple cubic, body centred cubic and face centred cubic, then the ratio of radii of the spheres in these system will be respectively.

(1) $\frac{1}{2}a : \sqrt{3}a : \frac{1}{\sqrt{2}}a$

(2) $\frac{1}{2}a : \frac{\sqrt{3}}{2}a : \frac{\sqrt{2}}{2}a$

(3) $1a : \sqrt{3}a : \sqrt{2}a$

(4) $\frac{1}{2}a : \frac{\sqrt{3}}{4}a : \frac{1}{2\sqrt{2}}a$

SS0085

18. An element crystallises in BCC structure. The edge length of its unit cell is 288 pm. If the density of the crystal is 7.2 g cm^{-3}, what is the atomic mass of the element?

(1) 51.8 (2) 103.6
(3) 25.9 (4) 207.2

SS0019

19. An element A crystallises in fcc structure. 200 g of this element has 4.12×10^{24} atoms. The density of A is 7.2 g cm^{-3} Calculate the edge length of the unit cell.

(1) 26.97×10^{-24} cm
(2) 299.9 pm
(3) 5×10^{-12} cm
(4) 2.99 cm

SS0021

20. Density of Li atom is 0.53 g cm^{-3}. The edge length of Li is 3.5 Å. Find out the number of Li atoms in an unit cell. ($N_A = 6.023 \times 10^{23}$), (M= 6.94 g mol^{-1})

(1) 2 (2) 8 (3) 4 (4) 6

SS0022

CLOSE PACKING OF IDENTICAL SOLID SPHERES

21. The coordination number of hexagonal closest packed (HCP) structure is

(1) 12 (2) 10 (3) 8 (4) 6

SS0023

22. The ABAB close packing and ABC ABC close packing are respectively called as

(1) hexagonal close packing(hcp) and cubic close packing (ccp)
(2) ccp and hcp
(3) body centred cubic (bcc) packing and hexagonal close packing (hcp)
(4) hcp and bcc

SS0024

23. The space occupied in BCC arrangement is

(1) 74% (2) 70% (3) 68% (4) 60.4%

SS0025

24. The vacant space in BCC unit cell is

(1) 32% (2) 10% (3) 23% (4) 46%

SS0026

25. The space occupied by spheres of equal size in three dimensions in both HCP and CCP arrangement is

(1) 74% (2) 70%
(3) 60.4% (4) 52.4%

SS0027

26. The empty space in the HCP and CCP is about

(1) 26% (2) 30% (3) 35% (4) 40%

SS0028

27. Which one of the following is not a close packing ?

(1) hcp (2) ccp
(3) bcc (4) fcc

SS0029

28. Close packing is maximum in the crystal lattice of

(1) Simple cubic
(2) Face centred
(3) Body centred
(4) Simple cubic and body centred

SS0030

29. Which of the following has HCP structure ?

(1) Al (2) Mg (3) Cu (4) Ni

SS0031

30. All noble gases crystallise in the CCP structure except

(1) Helium (2) Neon
(3) Argon (4) Krypton

SS0032

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31. If the coordination number of an element in its crystal lattice is 8, then packing is :
(1) FCC (2) HCP
(3) BCC (4) None of the above
SS0033

HOLES / VOIDS IN CRYSTALS

32. A tetrahedral void in a crystal implies that
(1) shape of the void is tetrahedral
(2) molecules forming the void are tetrahedral in shape
(3) the void is surrounded tetrahedrally by four spheres
(4) the void is surrounded by six spheres
SS0034

33. In a closest packed lattice, the number of octahedral sites as compared to tetrahedral ones will be
(1) Equal (2) Half
(3) Double (4) One fourth
SS0035

34. The coordination number of a cation occupying an octahedral hole is
(1) 4 (2) 6 (3) 8 (4) 12
SS0036

35. The size of an octahedral void formed in a closest packed lattice as compared to tetrahedral void is
(1) Equal (2) Smaller
(3) Larger (4) Not definite
SS0037

36. The coordination number of a cation occupying a tetrahedral hole is
(1) 4 (2) 6 (3) 8 (4) 12
SS0038

37. Number of tetrahedral voids per atom in a crystal lattice is :
(1) 1 (2) 2 (3) 4 (4) 8
SS0039

38. A compound contains P and Q elements. Atoms Q are in CCP arrangement while P occupy all tetrahedral sites. Formula of compound is :
(1) PQ (2) PQ_2 (3) P_2Q (4) P_3Q
SS0040

39. If 'Z' is the number of atoms in the unit cell that represents the closest packing sequence ABCABC---, the number of tetrahedral voids in the unit cell is equal to
(1) Z (2) 2Z (3) Z/2 (4) Z/4
SS0041

40. If a unit cell of fcc lattice is divided into eight identical small cubes, then
(1) Each small cube contains one octahedral void.
(2) Every alternate small cube contains one octahderal void.
(3) Each small cube contains one tetrahedral void.
(4) Every alternate small cube contains one tetraderal void.
SS0134

IONIC CRYSTAL

41. The limiting radius ratio for tetrahedral shape is
(1) 0 to 0.155 (2) 0.155 to 0.225
(3) 0.225 to 0.414 (4) 0.414 to 0.732
SS0042

42. For an octahedral arrangement the lowest radius ratio limit is
(1) 0.155 (2) 0.732 (3) 0.414 (4) 0.225
SS0043

43. If the radius ratio is in the range of 0.414 - 0.732 then the co-ordination number will be :
(1) 2 (2) 4 (3) 6 (4) 8
SS0044

44. In NaCl crystal r^+/r^- ratio is :
(1) 0.4 (2) 0.98 (3) 1.0 (4) 0.52
SS0045

45. Which one of the following statements is incorrect about rock salt type ?
(1) Na^+ occupy octahedral voids
(2) Na^+ and Cl^- ions have a co-ordination number of 6 : 6
(3) A unit cell of NaCl consists of four NaCl units
(4) All halides of alkali metals have rock-salt type structure
SS0046

46. In sodium chloride, Cl^- ions form ccp arrangement. Which site does Na^+ ions will occupy in this structure ?
(1) Cubical void (2) Tetrahedral void
(3) Octahedral void (4) Triangular void
SS0047

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47. The positions of Cl^- ions in NaCl structure are
(1) Corners of the cube
(2) Centres of faces of the cube
(3) Corners as well as centres of the faces of the cube
(4) Edge centres of the cube
SS0048

48. The number of NaCl units present in a unit cell of NaCl are
(1) 1 (2) 2 (3) 4 (4) 8
SS0049

49. The tetrahedral voids formed by ccp arrangement of Cl^- ions in rock salt structure are
(1) Occupied by Na^+ ions
(2) Occupied by Cl^- ions
(3) Occupied by either Na^+ or Cl^- ions
(4) Vacant
SS0050

50. The structure of MgO is similar to NaCl. The co-ordination number of Mg is
(1) 2 (2) 6 (3) 4 (4) 8
SS0051

51. The co-ordination number of Cs^+ and Cl^- ions in CsCl structure is
(1) 4 : 4 (2) 6 : 6 (3) 8 : 8 (4) 4 : 8
SS0052

52. A unit cell of CsCl consists of
(1) one CsCl unit (2) two CsCl units
(3) four CsCl units (4) eight CsCl units
SS0053

53. The NaCl structure can be converted into CsCl structure
(1) by application of high pressure
(2) by heating to 760 K
(3) both by heat and pressure
(4) the conversion is not possible
SS0054

54. TlCl has structure similar to CsCl. The co-ordination number of Tl^+ is
(1) 4 (2) 6 (3) 10 (4) 8
SS0055

55. The co-ordination number of Zn^{2+} and S^{2-} ions in the zinc blende (ZnS) type structure is
(1) 4 : 4 (2) 6 : 6 (3) 8 : 8 (4) 4 : 8
SS0056

56. The co-ordination number of calcium fluoride (CaF_2) type structure is
(1) 1 : 2 (2) 4 : 4 (3) 4 : 8 (4) 8 : 4
SS0057

57. The number of formula units in an unit cell of fluorite is
(1) 2 (2) 4 (3) 6 (4) 8
SS0058

58. 4 : 4 Co-ordination is found in
(1) ZnS (2) CuCl (3) Agl (4) All
SS0059

59. Antifluorite structure is derived from fluorite structure by
(1) Heating fluorite crystal lattice.
(2) Subjecting fluorite structure to high pressure.
(3) Interchanging the positions of positive and negative ions in the lattice.
(4) All of the above
SS0060

60. The radius of Na^+ is 95 pm and that of Cl^- ion is 181 pm. Hence the co-ordination number of Na^+ will be
(1) 4 (2) 6
(3) 8 (4) 12
SS0062

61. The ionic radii of Rb^+ and I^- are 1.46 and 2.16 Å respectively. The most probable type of structure exhibited by it is
(1) CsCl type (2) NaCl type
(3) ZnS type (4) CaF_2 type
SS0063

62. A solid XY has a bcc structure. If the distance of closest approach between the two ions is 173 pm, the edge length of the cell is
(1) 200 pm (2) $\frac{\sqrt{3}}{\sqrt{2}}$ pm (3) 142.2 pm (4) $\sqrt{2}$ pm
SS0065

63. If the distance between Na^+ and Cl^- ions in NaCl crystal is 'x' pm, what is the length of the cell edge?
(1) 2x pm (2) x/2 pm
(3) 4x pm (4) x/4 pm
SS0066

64. In a NaCl type crystal distance between Na^+ and Cl^- ion is 2.814 Å and the density of solid is 2.167 g cm^{-3} then find out the value of Avogadro constant.
(1) 6.05×10^{23} (2) 3.02×10^{23}
(3) 12.10×10^{23} (4) 9.6×10^{24}
SS0069

65. The density of crystalline sodium chloride is 5.85 g cm^{-3}. What is the edge length of the unit cell.
(1) 4.06×10^{-8} cm. (2) 1.32×10^{-14} cm
(3) 7.8×10^{-23} (4) 9.6×10^{-24}
SS0070

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66. A unit cell of sodium chloride has four formula units. The edge length of unit cell is 0.6 nm. What is the density of sodium chloride ?

(1) 7.60 g cm^{-3} (2) 1.80 g cm^{-3}

(3) 9.60 g cm^{-3} (4) 6.38 g cm^{-3}

SS0071

IMPERFECTIONS IN SOLIDS

67. At zero kelvin, most of the ionic crystals posses

(1) Frenkel defect (2) Schottky defect

(3) Metal excess defect (4) No defect

SS0072

68. As a result of Schottky defect

(1) there is no effect on the density

(2) density of the crystal increases

(3) density of the crystal decreases

(4) any of the above three can happen

SS0073

69. Schottky as well as frenkel defects are observed in the crystal of

(1) NaCl (2) AgBr (3) AgCl (4) $MgCl_2$

SS0074

70. Frenkel defect is generally observed in

(1) AgBr (2) AgI (3) ZnS (4) All of these

SS0076

71. Frenkel defect is found in crystals in which the radius ratio is :

(1) low (2) 1.3

(3) 1.5 (4) slightly less than unity

SS0077

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EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**3**	**4**	**2**	**3**	**2**	**1**	**3**	**2**	**1**	**1**	**1**	**1**	**3**	**2**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**2**	**4**	**1**	**2**	**1**	**1**	**1**	**3**	**1**	**1**	**1**	**3**	**2**	**2**	**1**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**3**	**3**	**2**	**2**	**3**	**1**	**2**	**3**	**2**	**3**	**3**	**3**	**3**	**4**	**4**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**3**	**3**	**3**	**4**	**2**	**3**	**1**	**1**	**4**	**1**	**4**	**2**	**4**	**3**	**2**
Que.	61	62	63	64	65	66	67	68	69	70	71				
Ans.	**2**	**1**	**1**	**1**	**1**	**2**	**4**	**3**	**2**	**4**	**1**				

EXERCISE-III (Analytical Questions) — Master Your Understanding

1. What is the % of volume occupied by atom in FCC

(1) 58% (2) 68%

(3) 74% (4) 88%

SS0120

2. Coordination number in HCP is :-

(1) 6 (2) 8

(3) 12 (4) 18

SS0121

3. The lattice structure of group I elements is

(1) FCC (2) BCC

(3) HCP (4) CCP

SS0122

4. Elements Na and Mg are Crystalizes in bcc and fcc respectively, then number of atoms of Na and Mg per unit cell are respectively :-

(1) 2, 4 (2) 9, 14

(3) 14, 9 (4) 4, 2

SS0123

5. In a face centred cubic lattice, atom A occupies the corner positions only and atom B occupies the face centre positions. If one atom of B is missing from one of the face centred points, the formula of the compound is :-

(1) A_2B_3 (2) A_2B_5

(3) A_2B (4) AB_2

SS0124

6. An alloy of copper, silver and gold is found to have copper constituting the fcc lattice. If silver atoms occupy the edge centres and gold is present at body centre, the alloy has a formula

(1) Cu_4Ag_2Au (2) Cu_4Ag_4Au

(3) Cu_4Ag_3Au (4) CuAgAu

SS0125

7. Expermentally it was found that a metal oxide has formula $M_{0.98}O$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be :-

(1) 7.01% (2) 4.08%

(3) 6.05% (4) 5.08%

SS0126

8. CsBr crystallises in a b.c.c. like ionic lattice. The unit cell length is 436.6 pm. Given that the atomic mass of Cs=133 and that of Br=80 amu and Avogadro number being $6.023\times10^{23}\,mol^{-1}$ the density of CsBr is :

(1) 42.5 g cm^{-3} (2) 0.425 g cm^{-3}

(3) 8.25 g cm^{-3} (4) 4.25 g cm^{-3}

SS0080

9. NaCl is doped with 10^{-4} mol% $SrCl_2$, the number of cation vacancies in one mole NaCl is –

(1) 6.02×10^{15} (2) 6.02×10^{16}

(3) 6.02×10^{17} (4) 6.02×10^{14}

SS0082

10. Which of the following statement is not correct ?

(1) Molecular solids are generally volatile

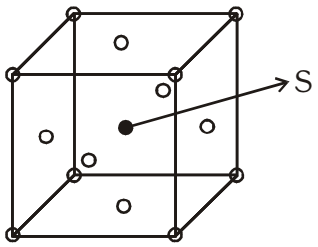
(2) The numbers of carbon atoms in an unit cell of diamond is 8

(3) The number of Bravais lattices in which a crystal can be categorized is 14

(4) The fraction of the total volume occupied by the atoms in a primitive cell is 0.48

SS0086

11. The site S in the adjoining fcc structure represents



(1) Octahedral void

(2) Tetrahedral void

(3) Triangular void

(4) Square void

SS0137

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12. Which of the following statement is correct about the hexagonal close packing.

(a) Coordination number is 12

(b) It has 74% packing effeciency.

(c) Tetrahedral voids of the second layer are covered by the spheres of the third layer

(d) It has 32% void

correct answer is :-

(1) a and b (2) a, b and c

(3) b, c and d (4) a,b,c and d

SS0138

13. Which of the following features are shown by quartz glass

(a) This is a crystalline solid

(b) This has definite Heat of fusion

(c) This is also called super cooled liquid

(d) Refractive index is same in all the directions.

Correct ans. is :-

(1) a and b (2) a, b and c

(3) c and d (4) a, b, c and d

SS0139

14. In which of the following arrangement octahedral voids are formed?

(a) BCC (b) FCC (c) HCP (d) SC

Correct ans. is

(1) b and d (2) a and c

(3) a, b and c (4) b and c

SS0140

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EXERCISE-III (Analytical Questions) **ANSWER KEY**

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Ans.	3	3	2	1	2	3	2	4	3	4	1	2	3	4	